

MEANING OF ECONOMICS

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Economics as being defined by many scholars as follows.

1. ALFRED MARSHALL - A study of man kind in an ordinary business of life.
2. ADAM SMITH - An inquiry into nature and causes of wealth of nations
3. JOHN STUART MILL - The practical science of production and distribution of wealth.
4. A.C. PIGOU - As a science of material welfare.

5. PROFESSOR LIONEL C. ROBBINS. The most accepted definition of economics. Defined economics as a science which studies human behaviours as a relationship between ends and scarce means which have alternative uses.
ENDS - means human desire or needs
SCARCE MEANS - limited available resources

NATURE AND SCOPE OF ECONOMICS

Economics is a social science subject because it studies human behaviour.

Economics is also regarded as science because it adopts the scientific method

~~It is a social science because it studies human behaviour.~~

BASIC CONCEPTS OF ECONOMICS

1. Want 2. Scarcity 3 Scale of preference 4 choice 5 opportunity cost.

* WANT - is the desire to own economic goods and services to satisfy human needs. Human wants are insatiable due to limited resources to satisfy them. The basic human needs in life includes food, clothing and shelter.

* SCARCITY - Scarcity means limited in supply, insufficient or not enough resources to satisfy human wants. It is the inability of human beings to satisfy all the things they want. The resources that will be used to satisfy want are limited because the ends are unlimited.

* SCALE OF PREFERENCE - The arrangement of wants in order of importance or priority. It is the list of numerous wants relative to their importance to the consumer.

* IMPORTANCE OF SCALE OF PREFERENCE

1. It helps in ranking of need according to their importance
2. It helps the consumer to be prudent in spending
3. It helps in identifying important needs
4. It helps in optimum allocation of resources
5. It maximizes consumer's satisfaction

* CHOICE - Choice is the selection of wants or needs from the numerous ends out of the various numbers of alternative available to the consumer.

* OPPORTUNITY COST - 1) is the cost of foregone alternative 2)

It is a cost that involve sacrificing one thing in order to obtain another. These are the wants that are left unsatisfied. Money cost the total amount of money spent in order to have a set of goods and services.

IMPORTANCE OF OPPORTUNITY COST

1. TO INDIVIDUALS

1. Wise choice 2. Efficient allocation of scarce resources

2. TO FIRMS

1. Rational decision about production process
2. It helps in the technique of production

3. TO THE GOVERNMENT

1. It helps in the preparation of budget
2. It helps in the allocation of resources.

WHY WE STUDY ECONOMICS

1. It helps in the allocation of scarce resources
2. It assist in the developmental programmes for the benefits of ^{CONSUMERS}
3. It helps in rational decision
4. It helps in the consumption of commodities
5. ~~It~~ faster maximization of profit

BRANCHES OF ECONOMICS

Economics is classified into two major groups, namely
1. Microeconomics and 2. Macroeconomics

* MICRO ECONOMICS - Deals with individual or smaller units of the economy such as household, firms and government

* MACRO ECONOMICS - It deals with aggregate or larger part of the economy such as national income, inflation, unemployment etc

OTHER BRANCHES OF ECONOMICS

1. Pure economics is concerned with laws and theories of economics
2. Applied economics is concerned with the application of laws
3. Mathematical economics is concerned with data analysis
4. Monetary economics is concerned with money and banking
5. Business economics - concerned with trade, business organization.
6. Development economics - concerned with planning.
7. Health economics - concerned with efficiency, effectiveness, values and behaviour in the production and consumption.
8. Engineering economics - concerned with determining from a number of technically acceptable alternatives, the one that provide the best economic result.

BASIC TOOLS FOR ECONOMIC ANALYSIS

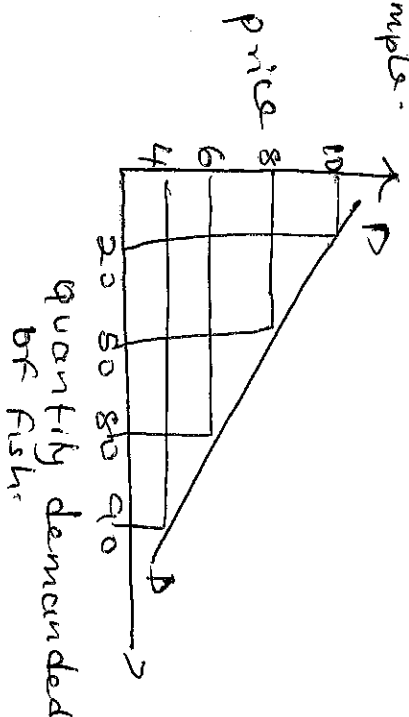
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1. TABLES - is the orderly arrangement of information

Example

Price	Number of bread demanded
100	1000
200	500
300	250
400	125

2. GRAPHS - is a diagram showing functional relationship between two variables. Example.



3. PIE CHART - It is a circle divided into sections each representing the value it represents. Example.

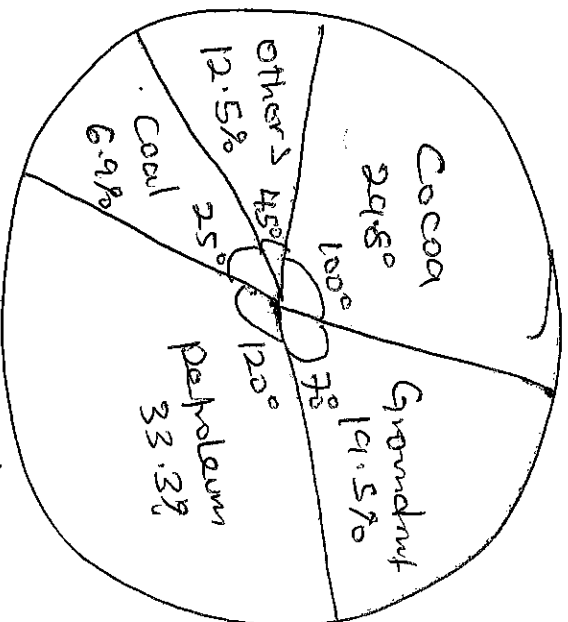
Value of the most important export in Nigeria.

Export product	Values in millions
Cocoa	200
Gumdamut	140
Petroleum	240
Coal	50
Others	90

Solution. 1. Add the total Value
ie $200 + 140 + 50 + 240 + 90 = 720$

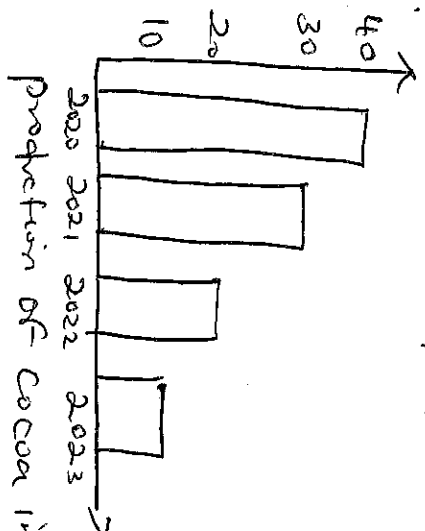
2. Workings in percentage and degree

Export product	Value in million	Workings in Percentage	Workings in degree
Cocoa	200	$\frac{200}{720} \times 100 = 27.7\%$	$\frac{200}{720} \times 360 = 100^\circ$
Gumdamut	140	$\frac{140}{720} \times 100 = 19.5\%$	$\frac{140}{720} \times 360 = 70^\circ$
Petrol	240	$\frac{240}{720} \times 100 = 33.3\%$	$\frac{240}{720} \times 360 = 120^\circ$
Coal	50	$\frac{50}{720} \times 100 = 6.9\%$	$\frac{50}{720} \times 360 = 25^\circ$
Others	90	$\frac{90}{720} \times 100 = 12.5\%$	$\frac{90}{720} \times 360 = 45^\circ$
Total	720	100%	360°



Pie chart showing the most important exports of Nigeria

4. BAR CHARTS - It is a graph made up of Rectangles which are equal in width and whose length are proportional to the quantities it represents. Example:
 Tippees

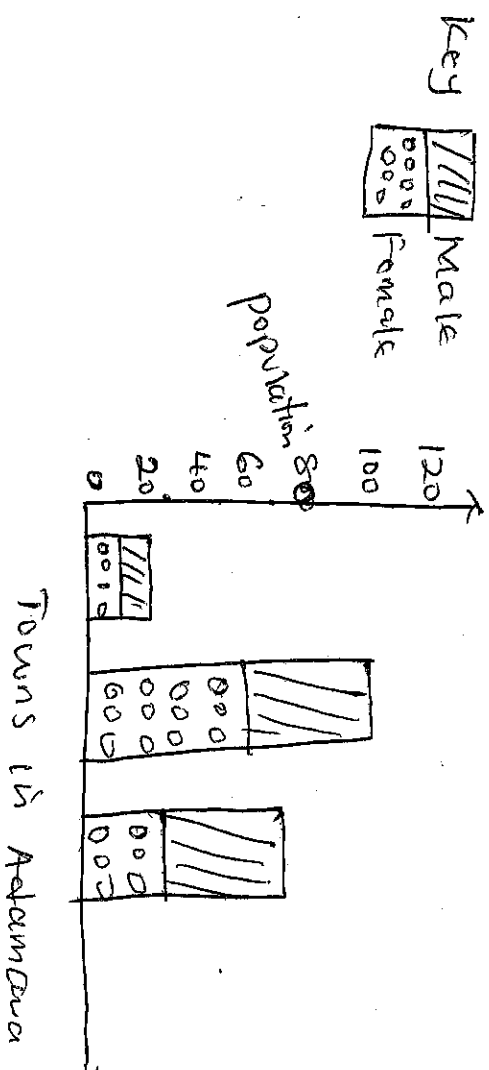


Production of Cocoa in Nigeria
2020 to 2023

Types of BAR CHARTS

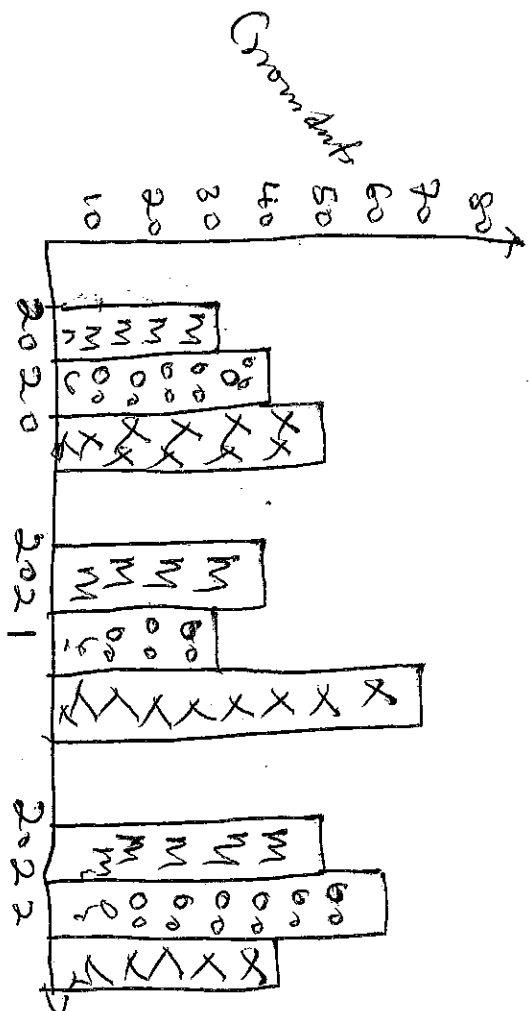
1. Component Bar chart - a chart that is used when it involved more than one variable. Example A population showing males and females in some towns in Adamawa state

Towns	Male	Female	Total
Mubi	10	10	20
HONG	40	60	100
SONG	30	40	70



3. MULTIPLE BAR CHART - is used when there is three or more variables example Nigerian Export of groundnut from Sokoto, Kano and Kaduna in 2020 to 2022

Towns	2020	2021	2022
Sokoto	30	40	50
Kano	40	30	60
Kaduna	50	70	40



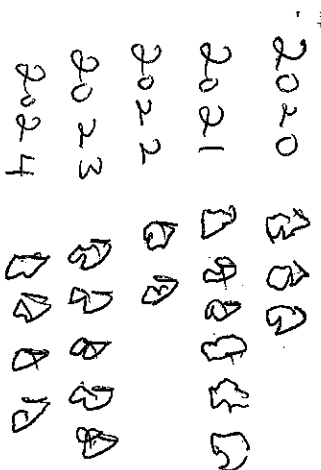
Key

M	Sokoto
O	Kano
X	Kaduna

Multiple bar chart showing Nigeria's towns Sokoto, Kano and Kaduna.

5 PICTOGRAMS - This is a charts of pictures representing an object. Example Chicken Consumed in Nigeria b/w 2020 to 2024

Year	Chickens in tonnes
2020	30
2021	60
2022	20
2023	50
2024	40



Each chicken represent 10 tonnes

6 HISTOGRAM - is a graphical representation of frequency distribution. It is made up of rectangles that have base on horizontal axis X and their frequencies on vertical axis Y. Example harvest of 60 tubers of yam for 8 days draw the histogram

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THEORY OF PRODUCTION

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Production involves all economic activities that is aimed at creation, transformation of raw material, creation of utility and distribution of goods and services to the final consumers.

Types of Goods

1. consumer goods 2. Producer or Capital goods.

* CONSUMER GOODS - This involves goods and services that satisfy consumer's immediate needs. e.g. milk, bread, rice, radio etc. It is also classified into 1. Durable consumers goods and 2. Non Durable consumers goods.

- DURABLE CONSUMERS GOODS - It is goods that consumer use over and over again e.g. radio, television, clothes, pots etc.

- NON-DURABLE CONSUMER GOODS - These are goods that are used at once for immediate satisfaction. e.g. eggs, bread, drugs etc.

* PRODUCER OR CAPITAL GOODS - These are goods used by individual firm in the production of another goods e.g. tools and machines, motors etc.

Types of Production

1. Direct production 2. Indirect production

* DIRECT PRODUCTION - It involves the production of goods and services for family consumption only. e.g. labour in the farm of the family.

* INDIRECT PRODUCTION - It involved production of goods and services in large scale for the purpose of commercializing it to consumers in the market.

Indirect Production Classified into THREE

1. Primary production 2. Secondary and 3. Tertiary production

* Primary Production - It is the extraction of raw materials from the land, sea and forest etc. e.g. agriculture, mining, fishing etc.

* Secondary Production - It is concern with changing the raw materials into semi and finished goods for consumption e.g. wheat change to flour and flour change to bread.

* Tertiary Production - It involves the provision of professional services through distribution to the final consumer e.g. doctors, lawyers, hair dresser, teacher etc.

Factors That Determined The Volume of Production

1. Amount of Capital - large amount of capital increase production.
2. Availability of raw materials - If raw materials are available it also increase production.
3. The market size - A large market size increase production.
4. Storage facilities - A large storage facilities also increase production.
5. Level of technology - A high level technology improve production.

FACTIONS OF PRODUCTION

These are agents that are combined in order to produce goods and services.

Types of FACTIONS OF PRODUCTION

1. Land 2. Labor 3 Capital 4 Entrepreneur.

x LAND - It is the free gift of nature that are used in production. Includes water, air, sunshine, atmosphere, forest etc. The reward for land is RENT
CHARACTERISTICS OF LAND

1. Land is immobile i.e. cannot be moved
2. Supply of land is fixed.
3. Land is a free gift
4. Land is subject to diminishing returns
5. Land is heterogeneous.

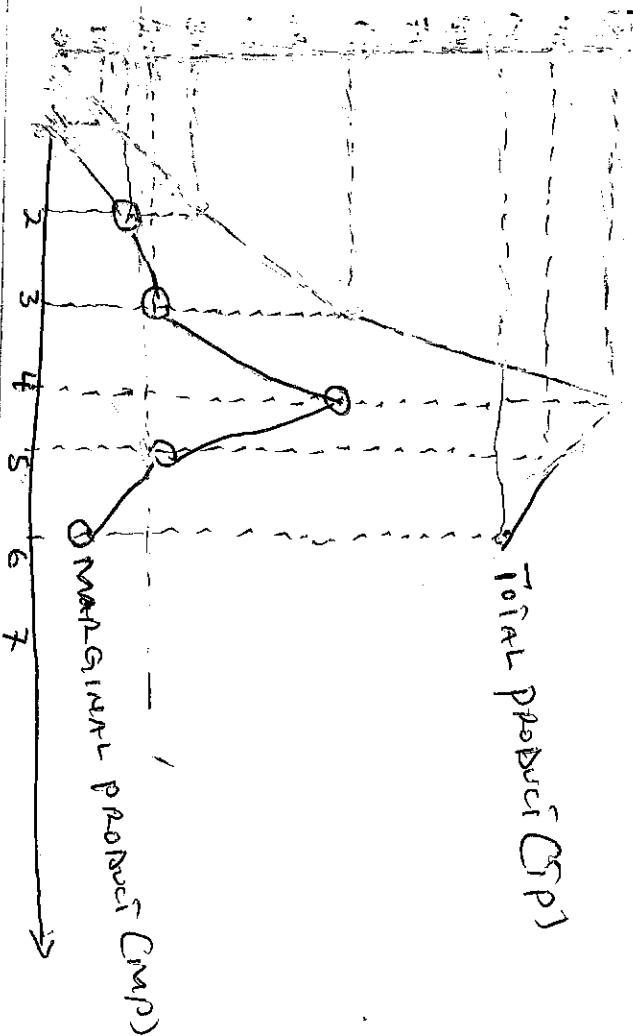
LAWS OF DIMINISHING RETURNS

The law states that as successive units of a variable factor e.g. labor is applied to a given fixed factor e.g. land, output will increase at first but will get to a point at which the addition of one more unit of the variable factor will result in less additional unit of output.

Illustration of law of diminishing returns of a farm of Mr A

Total factor Land	Variable factor Labour	Total output	Average output	Marginal output
1	1	10	10	-
1	2	30	15	20
1	3	60	20	30
1	4	120	30	60
1	5	100	20	20
1	6	90	15	10

From the table above one can depict that from the initial stage additional unit of variable factor increases the output from 10, 30, 60 and 120, but the application of the 5th unit of variable of labour decreases the output 100 from 120 thereby showing that the law of diminishing return has set in the 5th unit of labour. It can also be shown on a graph.



* LABOUR - is the mental and physical effort of human beings in the process of production. The reward for labour is WAGES AND SALARIES

1. Unskilled labour 2. Skilled labour 3. Types of Labour

* UNSKILLED LABOUR - is the type of labour that require little or no formal education. It is referred to as Brown collar jobs.

* SKILLED LABOUR - This is a type of labour that undergone training and obtain formal education. It is referred to as white collar jobs

CHARACTERISTICS OF LABOUR

1. Labour is mobile i.e. it can move from one place to another
2. Labour has feelings
3. Labour is a human factor
4. Labour is not fixed

* CAPITAL

- is a man-made asset is in the production of other wealth or goods and services. The reward for capital is INTEREST

CHARACTERISTICS OF CAPITAL

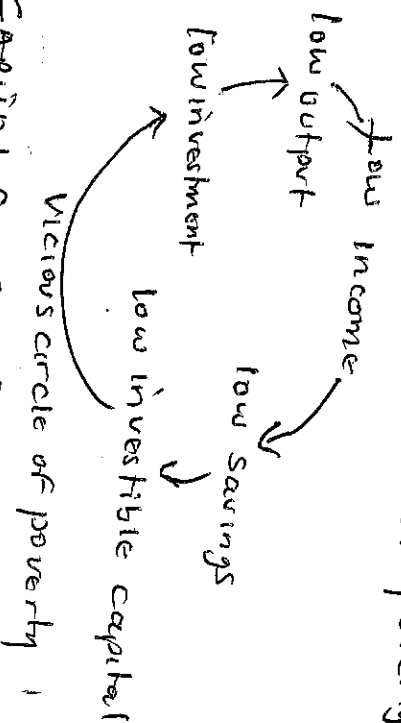
1. Capital is man made
2. Capital is durable e.g. machines
3. It ensures large scale production
4. It is subject to depreciation e.g. machines
5. It exists in different forms.

Types of Capital

1. Fixed Capital - It does not change in the process of production e.g. building machines
2. Circulating or Working Capital - It changes forms in the process of production e.g. raw materials, water, fuel
3. Current or liquid Capital - This capital are needed for day to day running of production e.g. money
4. Social Capital - This are assets provided by government e.g. water electricity, roads etc.

CAPITAL FORMATION OR ACCUMULATION

It means increasing a country's stock of real capital i.e. increasing the investment in form of fixed assets. For capital to accumulate, a country must reduce consumption and increase savings for investment. A country with low savings will lead to low investment and result to what economist referred to as Vicious Circle of poverty i.e. capital



CAPITAL CONSUMPTION

Capital Consumption means using up of existing capital stock and not replacing worn-out capital goods used in production.

* ENTREPRENEUR

- A firm or individual that co-ordinate all the factors of production. The reward for ~~capital~~ Entrepreneur is PROFIT

1. Risk Bearer
2. Organizing
3. Decision making
4. Controls other factors

WHY IS EXTERNAL ECONOMIES

2. 4-

These are the benefits a firm derives from concentration of industries in a particular area e.g. roads, pipe, bonewater, electricity.

~~WHY IS DISADVANTAGE~~
WHY IS EXTERNAL DISADVANTAGES

The disadvantages a firm experience when the activities of one or more industries increases the cost of production or output.

DIVISION OF LABOUR

It's the breaking down of production process into different parts whereby each operation is carried out by one person or group of persons.

SPECIALIZATION

It's the concentration of productive efforts of an individual in a given aspect in a particular line of production.

TYPES OF SPECIALIZATION

- 1 Specialization by process - a situation in which production process is divided into different operation for each worker to specialize in one particular process
 - 2 Specialization by sex e.g. male and female
 - 3 Specialization by product - one particular product process
 - 4 Geographical specialization - based on the natural endowment
- Limitation To Division of Labour
- 1 Size of the market
 - 2 Level of technology
 - 3 Availability of capital
 - 4 Government policy
 - 5 Nature of the product e.g. banking, printing

SMALL AND LARGE FIRMS

WHY IS A FIRM - is an independent administered business unit which is capable of carrying out production and distribution.

ECONOMIC OF SCALE OR SCALE OF PRODUCTION

Economic of scale is the growth of firms as a result of the expansion of the volume of productive capacity resulting in the increase in output and decrease in its cost of production per unit cost of the output.

TYPES OF ECONOMIES OF SCALE

- 1 Internal economies of scale
- 2 External economies

* INTERNAL ECONOMIES - The advantage which a firm derives as a result of its increase in size and expansion of its output.

* EXTERNAL ECONOMIES - are the benefits a firm derives from concentration of industries in a particular area.

DEMAND

(1)

Demand is the amount of economic goods and services that the consumer is willing and able to buy at a particular price in a particular period of time.

LAW OF DEMAND

1. The first law of demand states that the higher the price the lower the quantity demanded by consumers for normal goods. Inverse relationship.
2. For exceptional or abnormal goods, the higher the price the higher the quantity demanded. Direct relationship.

DEMAND SCHEDULE

A demand schedule is a table showing the quantity demanded by a consumer and at a different price.

MARKET DEMAND

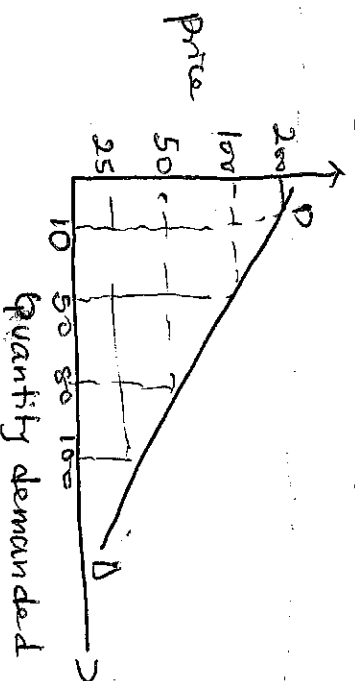
It is the total quantity of goods bought by consumers for a period of time in a market.

FACTORS THAT AFFECT DEMAND

1. Price 2. Price of other goods 3. Income of consumers 4. Period of festivals 5. Population 6. Changes in taste and fashion

DEMAND CURVE

It is a curve showing the relationship between price and quantity demanded of a goods.

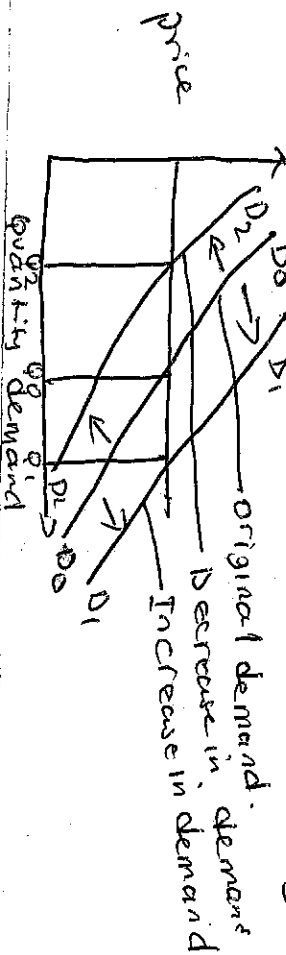


TYPES OF DEMAND

1. Derived Demand - It occurs as a result of demand for other goods e.g. labour as a factor of production.
2. Joint or complementary Demand - It occurs when two or more commodities are needed at the same time e.g. petrol and car.
3. Competitive Demand - When two or more commodities are fairly close substitutes to each other e.g. bournvita and milko.
4. Composite Demand - When a commodity is required for two or more purposes e.g. wood.

CHANGE IN DEMAND

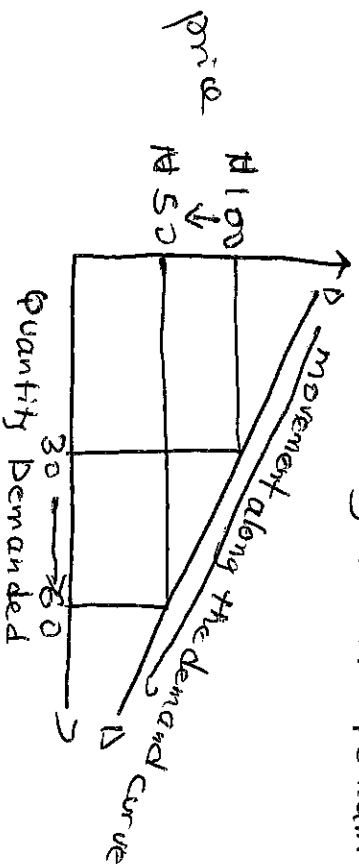
A change in demand is the bodily shift of demand curve either to the left, meaning decrease or contraction or to the right meaning increase.



CHANGE IN QUANTITY DEMANDED

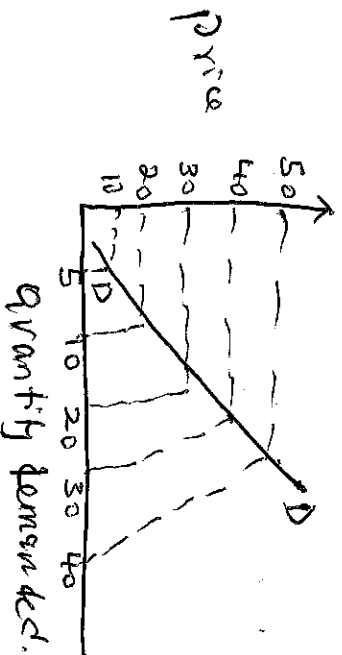
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Change in quantity demanded is the movement along the demand curve caused by price while other factors affecting demand remain constant.



EXCEPTIONAL OR ABNORMAL DEMAND CURVE

In this case the higher the price the higher the quantity demanded. It does not follow the rules of normal demand curve. e.g. diamond, golds



CAUSES OF EXCEPTIONAL DEMAND CURVE

1. Articles of ostentation - goods that have luxury and prestige like gold.
2. Necessity - also known as Giffen or inferior goods. e.g. Gari, sugar. No matter the increase in price, consumer will still buy the goods.
3. Future expectation - if consumer believes that the prices of goods will rise in the future, he will buy at the prevailing high price.
4. Rare goods - These are unique goods such as paintings and antiques that are not common.

PRICE ELASTICITY OF DEMAND

It is the degree of responsiveness of the quantity demanded of a commodity to change in the price of the commodity. It measures how consumer reacts to change in price as a result of quantity demanded. Formula:

$$PED = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}} \times 100$$

TYPES OF PRICE ELASTICITY OF DEMAND

1. Elastic Demand - If the percentage change in price leads to a greater change in quantity demanded. It is fairly elastic. It is always greater than one. Example: The price and quantity of fish or MRA.

Price	Quantity demanded
₹40	160
₹50	80

1. Find the percentage change in quantity
2. Find the percentage change in price
3. Find the coefficient of elasticity

$$\text{Solution: } \frac{\text{Percentage change in quantity}}{\text{Change in quantity}} \times \frac{100}{\frac{\text{Old quantity}}{1}} = \frac{160 - 80}{80} \times \frac{100}{1} = \frac{80}{80} \times 100 = 100$$

$$\frac{80}{80} \times 100 = 100$$

2. Percentage change in price = $\frac{\text{change in price} \times 100}{\text{old price}} = \frac{50-40}{40} \times 100 = 25\%$ ③

$\frac{10}{40} \times 100 = 25\%$

3. coefficient of elasticity = $ED = \frac{50\%}{25\%} = 2$. It is elastic demand because its greater than one.

⑤ INFELASTIC DEMAND: If the percentage change in price is less than percentage change in quantity demanded. It is always less than one.

Example: price of bread for MRB.

price	quantity demanded
Rs 5	20
Rs 2	16

1. Find the percentage change in quantity
2. Find the percentage change in price
3. Find the coefficient

Solution 1: percentage change in quantity = $\frac{\text{change in quantity demanded} \times 100}{\text{old quantity}}$

$\frac{20-16}{20} \times 100 = \frac{4}{20} \times 100 = 20\%$

2. percentage change in price = $\frac{\text{change in price} \times 100}{\text{old price}} = \frac{7-5}{5} \times 100 =$

$\frac{2}{5} \times 100 = 40\%$

3. coefficient of elasticity = $\frac{20\%}{40\%} = \frac{1}{2}$ or 0.5 is inelastic demand because it is less than one.

⑥ UNITARY ELASTIC DEMAND: If the percentage change in price equal the percentage change in quantity demanded. It is always equal to one.

INCOME ELASTICITY OF DEMAND

Is the degree of responsiveness of demand to changes in income of a consumer. How changes in income of consumers affect the quantities demanded by such consumer. Formula

$IED = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$

Percentage change in income

TYPES OF INCOME ELASTICITY OF DEMAND

1. Positive income elasticity of demand - An increase in income will lead to increase in the quantity demanded.

2. Negative income elasticity of demand - An increase in income of consumer will lead to a decrease in the quantity demanded. eg Gari

Example of Income elasticity: A monthly income of a housekeeper increased from Rs 100 to Rs 125 as a result of promotion. 1. Calculate the coefficient of his income 2. Is the demand elastic or inelastic 3. What kind of goods is sugar to the consumer. Solution:

Income		quantity demanded	
Old	New	Old	New
100	125	200	300

1. Percentage change in quantity demanded.

$$= \frac{\text{New quantity} - \text{old quantity}}{\text{old quantity}} \times \frac{100}{1}$$

$$= \frac{300 - 200}{200} \times \frac{100}{1} = \frac{100}{200} \times \frac{100}{1} = 50\%$$

2. Percentage change in income = $\frac{\text{New income} - \text{old income}}{\text{old income}} \times \frac{100}{1} =$

$$= \frac{125 - 100}{100} \times \frac{100}{1} = \frac{25}{100} \times \frac{100}{1} = 25\%$$

3 coefficient of income elasticity = $\frac{50\%}{25\%} = 2.0$ It is elastic because it is greater than one.

4. The kind of goods ~~low~~ Sugar is to the consumer is normal goods. because income increases and the consumption of sugar also increases.

CROSS ELASTICITY OF DEMAND

It is the degree of responsiveness of demand for a commodity to a change in the price of other commodity. It is mainly concerned with close substitute as well as complementary goods. For example the demand for butter will increase if there is an increase in the price of margarine.

It is measured as; $CED = \frac{\text{Percentage change in quantity demanded of commodity X}}{\text{Percentage change in the price of Y}}$

Commodity	Change in price		Commodity	Change in quantity demanded	
	Original price	New price		Original quantity	New quantity
Maltina	50	80	Maltina	200	300
Close up	50	100	Macean	120	150

Example:

1. Calculate the Cross elasticities of demand for maltina and maltina.

2. Close up and macean 3 are their elasticities elastic or inelastic.

Solution: Cross elasticity of demand for maltina and maltina.

Let X = maltina and Y = Maltina

1. Percentage change in quantity demanded of maltina (X)

$$\frac{\text{New quantity} - \text{original quantity}}{\text{original quantity}} \times \frac{100}{1}$$

$$\frac{300 - 200}{200} \times \frac{100}{1} = \frac{100}{200} \times \frac{100}{1} = 50\%$$

2. percentage change in the price of maltina (Y)

(5)

$$= \frac{\text{New price} - \text{Original price}}{\text{Original price}} \times \frac{100}{1} = \frac{80 - 50 \times 100}{50} = \frac{30}{50} \times 100 = 60\%$$

Cross elasticity of maltonic and maltina

$$= \frac{50\%}{60\%} = 0.83\%$$

② Cross elasticity of demand for macleen and close up

Let X = Macleen and Y = Close up

= Percentage change in quantity demanded of macleen (X)

$$= \frac{\text{New quantity} - \text{Original quantity}}{\text{Original quantity}} \times \frac{100}{1} = \frac{150 - 120 \times 100}{120} = \frac{30}{120} \times 100 = 25\%$$

③ Percentage change in the price of Close up (Y) =

$$\frac{\text{New price} - \text{Original price}}{\text{Original price}} \times \frac{100}{1} = \frac{60 - 50 \times 100}{50} = \frac{10}{50} \times 100 = 20\%$$

Cross elasticity of demand for macleen and Close up:

$$= \frac{25\%}{20\%} = 1.25$$

④ The cross elasticity for maltina and maltonic is inelastic, i.e. 0.83 is less than one.

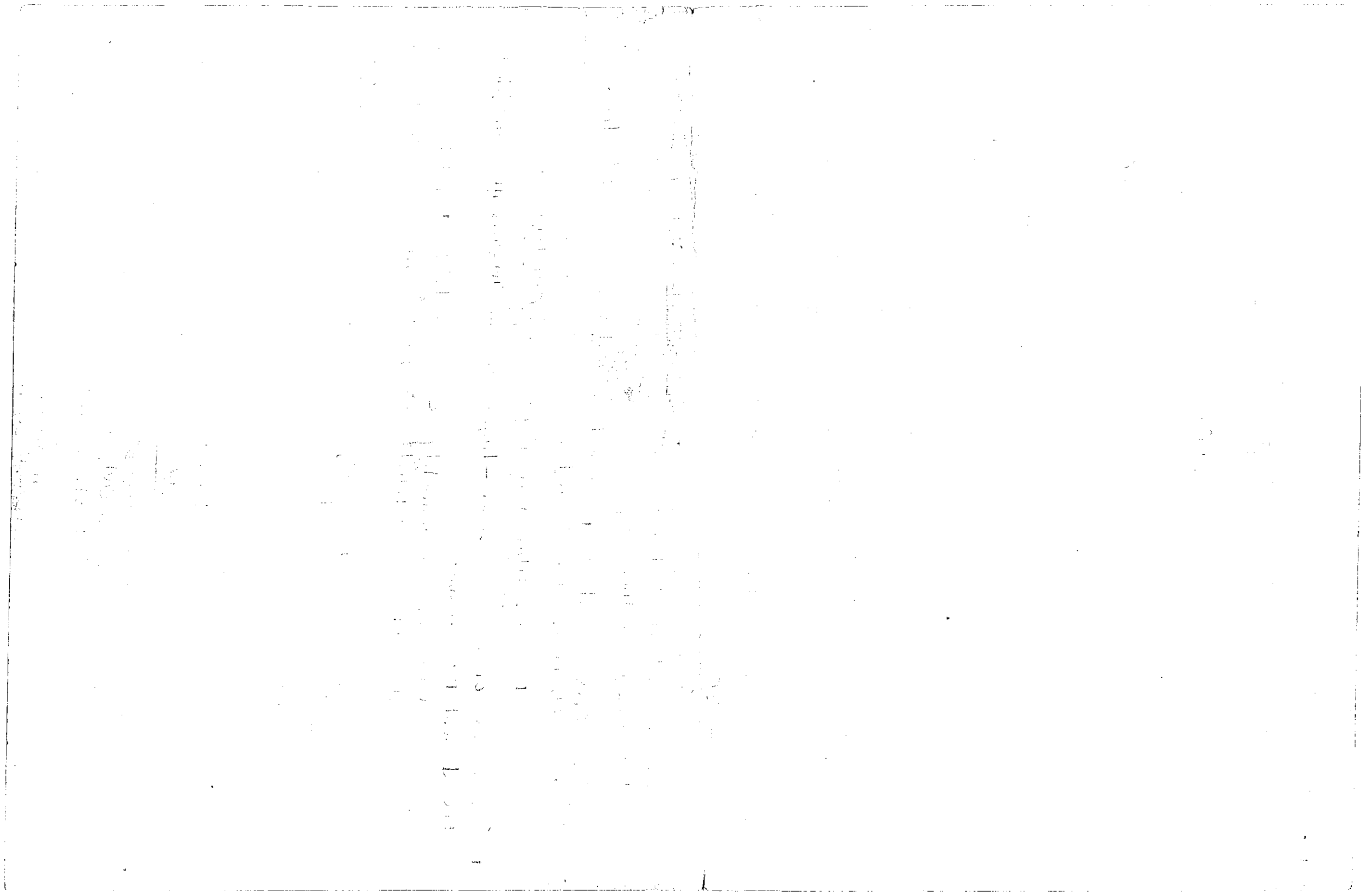
⑤ The Cross elasticity for macleen and Close up is elastic, i.e. 1.25 is greater than 1.

FACTORS THAT AFFECT ELASTICITY OF DEMAND

1. The income of the consumer - higher income means, more inelastic the demand for consumer.
2. Close substitute of a commodity - close substitutes tend to have high price elasticity of demand while non close substitute have inelastic demand.
3. Nature of commodity - necessity goods have inelastic demand, luxury goods have elastic demand.
4. One's habit - a high habit for goods tend to be inelastic demand while non habit goods are elastic.
5. Number of uses to which a commodity is put - if the commodity is put into several uses it is tend to have elastic demand vice versa.

IMPORTANCE OF ELASTICITY OF DEMAND

1. Increase revenue - if the goods is inelastic.
2. Determination of maximum output - It help producer to determine output.
3. Determination of cross elasticity - It help producer to determine which goods to produce.
4. Imposition of taxes by government - It help government to determine which goods to be taxed more.



THEORY OF CONSUMER BEHAVIOUR

It is concerned with how a consumer maximizes his satisfaction from goods and services by dividing his limited amount of income.

UTILITY-1

Is the satisfaction derived by consumer from consumption of goods and services

TYPES OF UTILITY-1

1 Time Utility 2 Form Utility 3 Place Utility

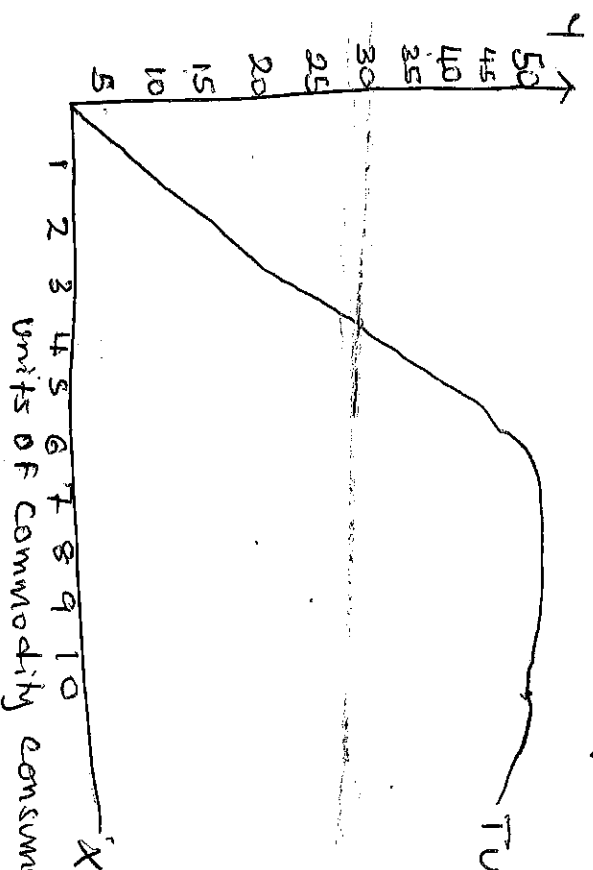
* TIME UTILITY-1 - It is the ability of a commodity or service to satisfy a consumer's want at a particular time

* FORM UTILITY-1 - This is the transformation of a commodity from one form to another to satisfy consumer's want e.g. flour to bread.

* PLACE UTILITY-1 - This is goods that satisfy consumer's wants if they are shifted to another place e.g. USA to Nigeria

CONCEPTS OF TOTAL, MARGINAL AND AVERAGE UTILITY

* TOTAL UTILITY-1 - This is the total amount of satisfaction a consumer derives from consumption of a particular commodity



* MARGINAL UTILITY-1 - It is additional satisfaction a consumer derives from consumption of additional unit of a particular commodity. It is calculated as: $MU = \frac{\text{Change in } TU}{\text{Change in consumption}}$

Change in consumption

* AVERAGE UTILITY-1 - This is the amount of satisfaction a consumer derives from the consumption of a unit of commodity. It is calculated as $AU = \frac{TU}{\text{Quantity consumed}}$

ILLUSTRATION OF TOTAL, MARGINAL AND AVERAGE UTILITY

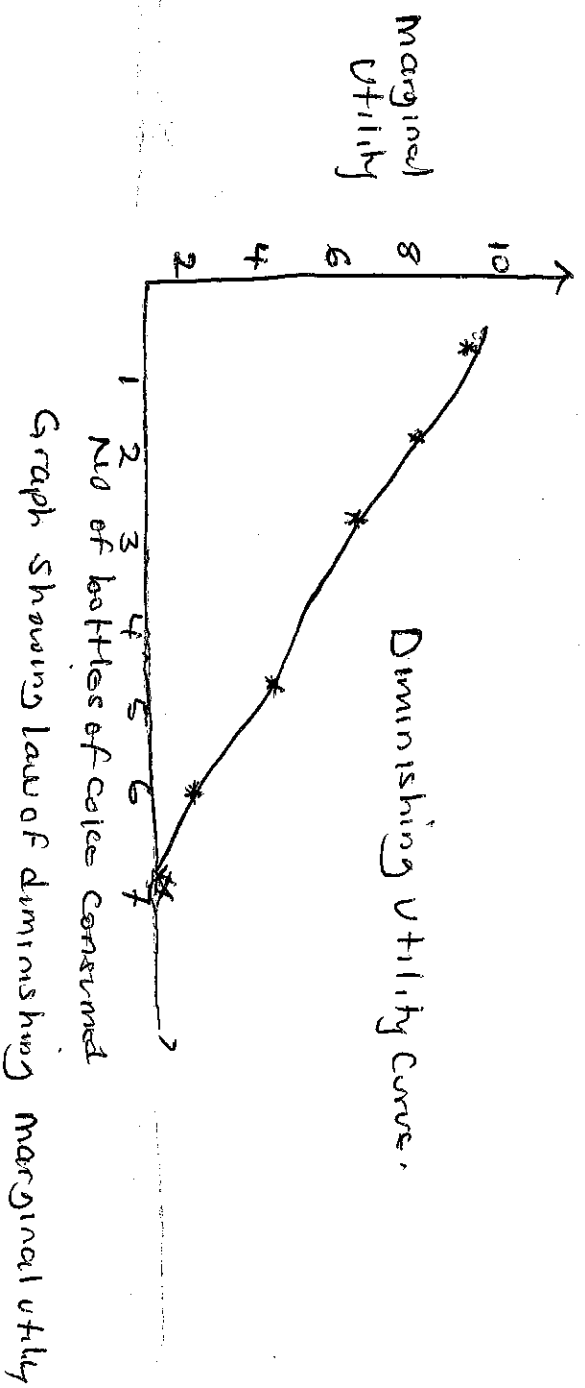
Quantity of goods consumed	Total Utility	Average Utility and Marginal Utility
1	10	10
2	14	7
3	18	6
4	20	5
5	25	5
6	30	5

LAW OF DIMINISHING MARGINAL UTILITY (2)

This law states that as a consumer consumes more and more units of a particular commodity, utility will increase upto certain point, when decrease in satisfaction will set in as a result of continuous consumption of the same commodity.

ILLUSTRATION OF LAW OF DIMINISHING MARGINAL UTILITY

No of bottles of Colce Consumed	Total Utility	Average Utility	Marginal Utility
1	9	9	-
2	16	8	9
3	24	8	8
4	30	7.5	6
5	34	6.8	4
6	36	6	2
7	36	5.1	0



UTILITY MAXIMIZATION

It is the equilibrium point of a consumer. A point where a consumer derives maximum satisfaction where his marginal utility equates the price of commodity consumed. A rational consumer maximizes utility where:

$$\frac{MU_X}{P_X} = \frac{MU_Y}{P_Y} \quad \text{where: } MU_X \Rightarrow \text{Marginal Utility of Commodity X}$$

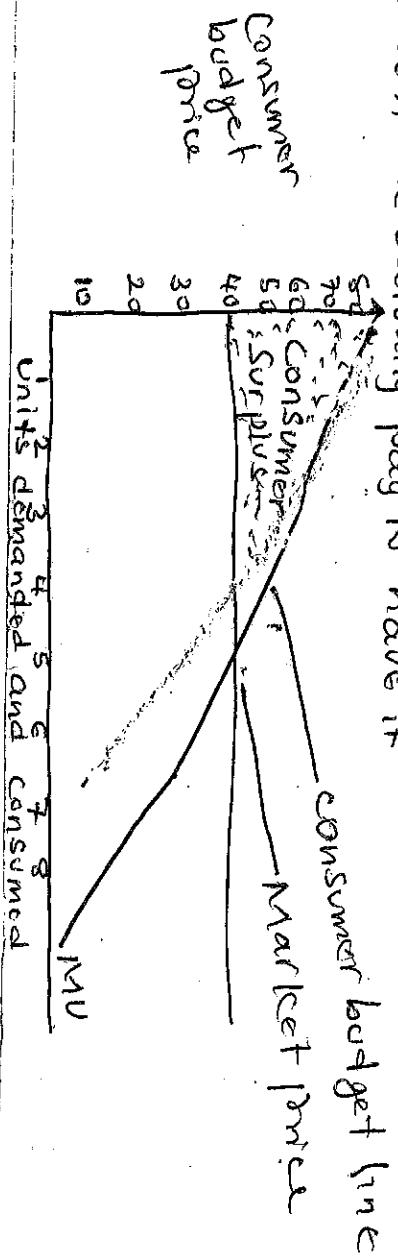
$$P_X \Rightarrow \text{Price of Commodity X}$$

$$MU_Y \Rightarrow \text{Marginal Utility of Commodity Y}$$

$$P_Y = \text{Price of Commodity Y}$$

CONSUMER SURPLUS

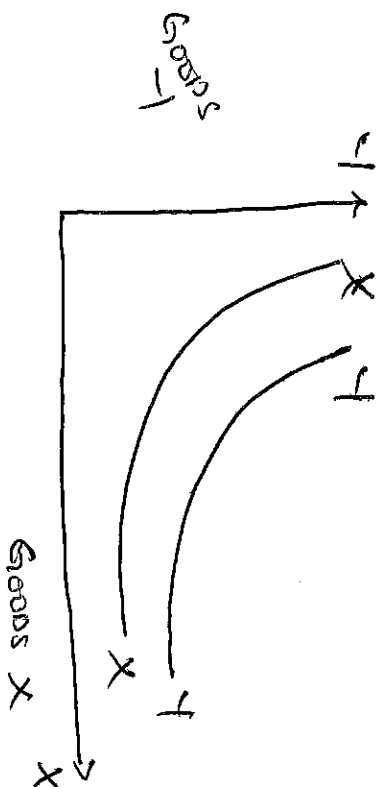
A consumer surplus is the difference between the amount a consumer budgeted to pay for a commodity based on anticipated level of satisfaction and the amount he actually pay to have it.



INDIFFERENCE CURVE

(3)

Is a curve that represents all the combination of goods that give the same satisfaction to the consumer.



From the graph, the higher the level of indifference, the higher the satisfaction. $\downarrow$ is greater than X in terms of satisfaction.

PROPERTIES OF INDIFFERENCE CURVE

1. Indifferent curve do not intersect, it never crosses each other
2. Indifferent curve that move rightward shows higher level of satisfaction

MARGINAL RATE OF SUBSTITUTION (MRS)

The marginal rate of substitution is the amount of one good that a consumer is willing to sacrifice in exchange for some amount of another goods for the same level of satisfaction.

THEORY OF COST

It is a concept used to explained the relationship that exist between various cost of production.

COST OF PRODUCTION

Is the sum total of all the payments made for the factors of production e.g Land, Labor, Capital and Entrepreneur.

THE COST CONCEPTS TYPES

1. Fixed cost (FC)
2. Variable cost (VC)
3. Marginal cost (MC)
4. Average cost (AC)
5. Total cost (TC)

* **FIXED COST (FC)** - It is a cost that does not change in the process of production. e.g building, factory, motor vehicle, machinery etc.

Formula for Fixed cost: $FC = TC - VC$ OR $TC = AFC \times \text{Quantity produced}$

* **VARIABLE COST (VC)** - A cost that increases when the producer increases its output. e.g fuel, raw materials, labor etc. Formula for variable cost: $VC = TC - FC$

* **MARGINAL COST (MC)** - is the extra additional cost of producing one more unit of output. Example if the cost of producing 20 litres of groundnut oil is Rs 10 and it cost Rs 11 to produce 21 litres of groundnut oil the difference is known as Marginal cost. Formula for calculating Marginal cost: $MC = \frac{\text{Change in TC}}{\text{Change in output}}$

Change in output.

* AVERAGE COST (AC) - is the cost per unit of an output. It is the firms total cost divided by the output. It is further divided to Average fixed cost (AFC) and Average variable cost (AVC). The formula for calculating Average cost is:

$$AC = \frac{\text{Total cost (TC)}}{\text{Total output}} \text{ or } AFC = AFC + AVC.$$

* AVERAGE VARIABLE COST (AVC) - is $AVC = \frac{TVC}{TQ}$ or $AFC - AVC$

* AVERAGE FIXED COST (AFC) - Total fixed cost divided by the output. $AFC = \frac{TFC}{\text{output}} = AFC - AVC$

* TOTAL COST - It is the sum of fixed cost (FC) and variable cost (VC) incurred in the production process. $TC = FC + VC$
It is made up of TFC and TVC.

ILLUSTRATION OF ALL THE COST CONCEPTS USED ABOVE

Unit of output	Total fixed cost (TFC)	Total variable cost (TVC)	Total cost (TC)	Average Total cost (ATC)	Average Variable cost (AVC)	Average Fixed cost (AFC)	Marginal Cost (MC)
1	20	12	32	32	12	20	-
2	20	14	34	17	7	10	2
3	20	16	36	12	5.3	6.6	2
4	20	18	38	9.5	4.5	5	2
5	20	20	40	8	4	4	4
6	20	22	42	7	3.6	3.3	2
7	20	24	44	6.3	3.4	2.8	2

Supply

X

①

It is the quantity of commodity that the producer is willing to put onto the market for sale at the market price over a period of time.

LAW OF SUPPLY

It states that the higher the price, the higher the quantity supplied by the producer. Vice versa. Supply has a direct relationship with price.

SUPPLY SCHEDULE

A table showing the quantity supplied and at different prices.

Types of supply schedule

1. Individual supply schedule 2. Market supply schedule.

* INDIVIDUAL SUPPLY SCHEDULE - It is the quantity supplied by an individual at different prices for sale at a particular period of time.

Example: A baker supply of bread in a week.

Price per loaf	No. of quantity supplied.
100	50
80	40
60	30
40	20
20	10

From the above table, when the price was ₹100, 50 loaves of bread was supplied, but when the price was ₹20, only 10 loaves were supplied.

* MARKET SUPPLY SCHEDULE - It is the quantity of commodity that all producers are willing to supply in the market at various prices. Example: Supply of tomatoes by, Mr Peter, Mr Adamu, Mr John.

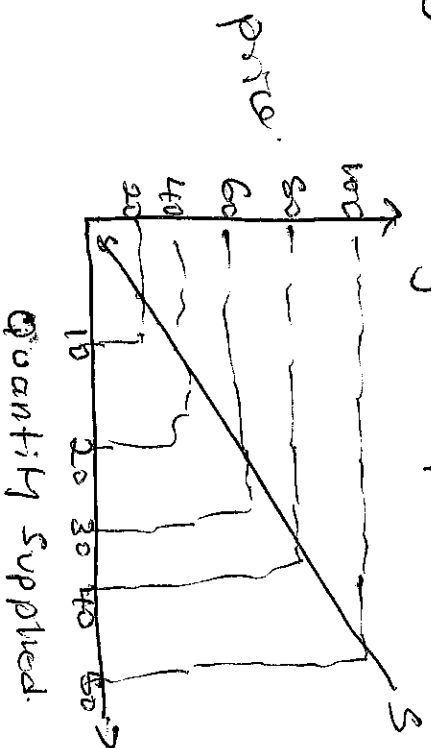
Price per basket	Quantity supplied.			Total quantity supplied.
	Mr Peter	Mr Adamu	Mr John	
100	50	80	70	200
80	40	70	50	160
60	30	60	30	120
40	20	50	20	90
20	10	40	10	60

SUPPLY CURVE

X

(2)

Is a graph showing the functional relationship between price and quantity supplied. It is the representation of the Supply Schedule diagrammatically. Example



FACTORS THAT AFFECT SUPPLY

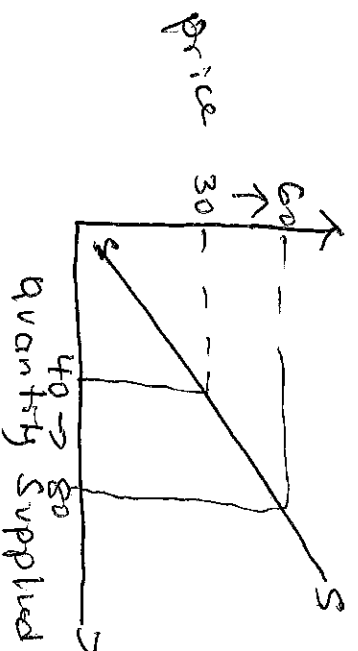
1. Price - When prices of goods are higher, suppliers increase supply in the market.
2. Cost of production - A higher cost of production means reduction in the quantity supplied by suppliers.
3. Weather - A favourable weather especially in agriculture tends to increase supply for that product.
4. Taxation - Higher taxes tend to discourage the producer from producing more.
5. Technology - A highly technological production method will increase speed of production of goods for consumption and in abundance.
6. Prices of other commodities - If the price of a commodity increases supply will increase vice versa.

TYPES OF SUPPLY

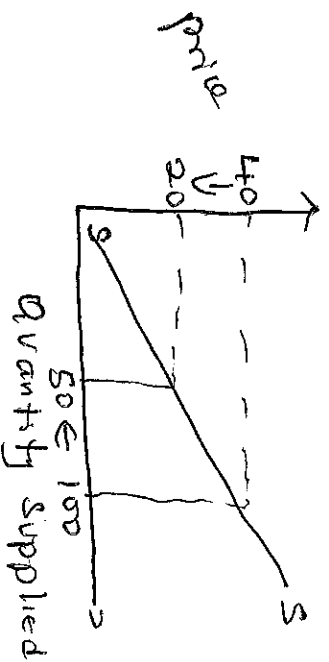
1. Composite supply - A situation where a commodity can serve two or more purposes. E.g. Crude oil. The supply of petrol and kerosene from the crude. The increase in supply of one will affect the supply of the other.
2. Joint or complementary supply - This is where two or more commodities are produced and supplied from one source. An increase in supply of one will lead to an increase in the supply of the other. E.g. petrol and kerosene from crude oil.
3. Composite supply - When a commodity is supplied for the satisfaction of a particular want. It is the supply of two or more commodities that serve as substitutes. E.g. Viva and Klein.

CHANGE IN QUANTITY SUPPLIED

It means a movement from one point to another on a supply curve. It is caused by a result of price increase or decrease while other factors remain constant.



③ X
Increased in the Quantity Supplied. i.e. if the Quantity increase as a result of increase in price. i.e. expansion of Supply.

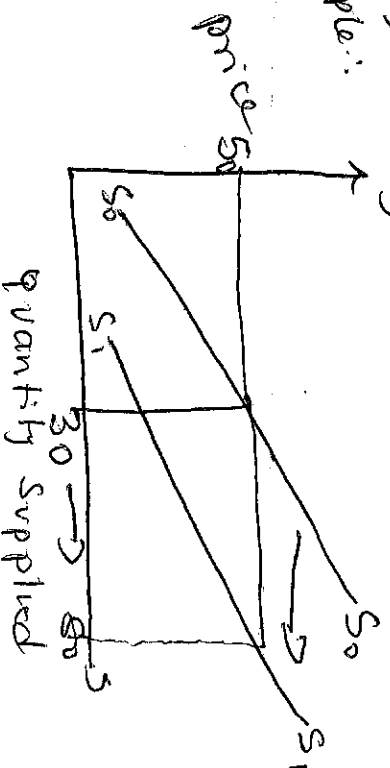


Decrease in Quantity Supplied as a result of decrease in Price. i.e. Contraction of Supply Curve.

CHANGE IN Supply

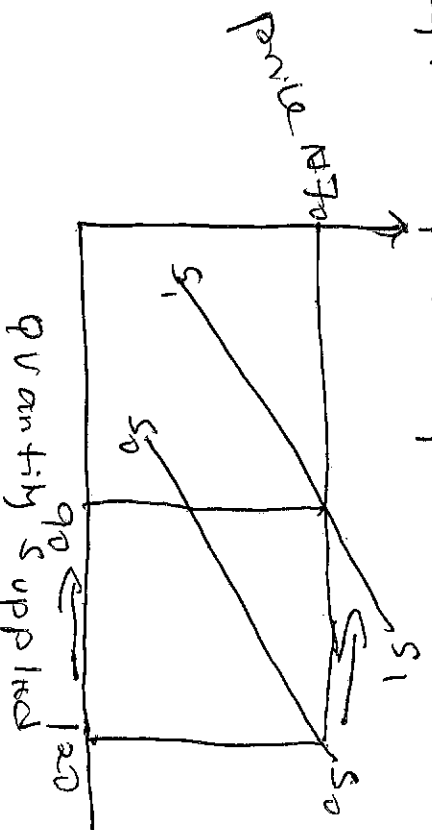
It is the bodily shift of the supply curve from one entire position to another. It is caused by other factors affecting supply with the exception of price. It can be increase or decrease.

INCREASE IN Supply - It is the shift of the supply curve to the right, showing that at that old price more of the goods is supplied.



The supply curve shift from S_0 to S_1 , at the old price of 50 and quantity supplied increase from 30 to 50.

DECREASE IN Supply - is a situation where the supply curve shift to the left, showing that at old price less of the goods will be supplied. All factors affecting supply are responsible for the shift with the exception of price.



The supply curve shift from S_0 to S_1 , at the old prevailing price of 70, and reduction in quantity supplied from 120 to 90.

ELASTICITY OF SUPPLY

X

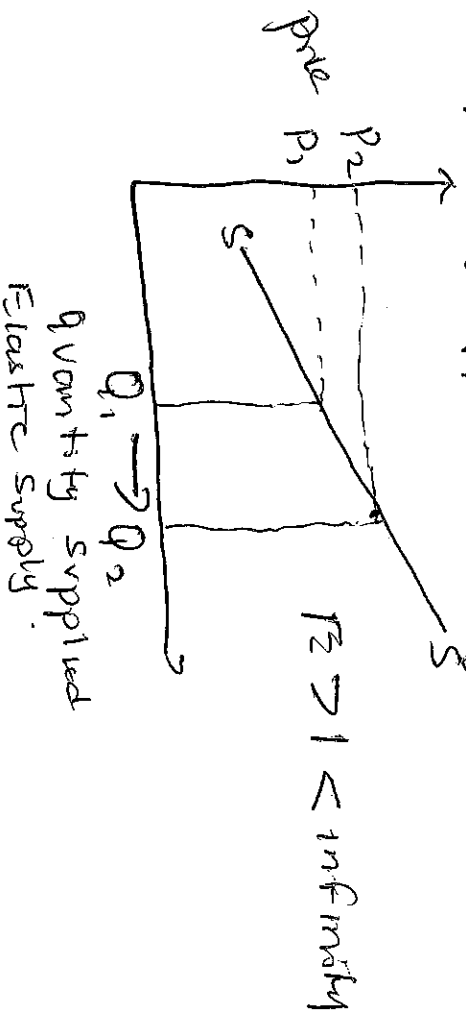
(4)

It is the degree of responsiveness of the quantity supplied of a goods to change in the price of a commodity. It measures the extent of the quantity of a goods supplied by a producer as a result of little change in price of the goods. Formula

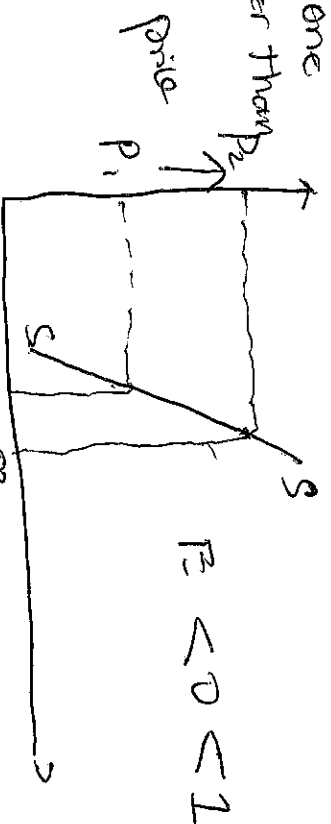
$$ES = \frac{\text{Percentage change in supply}}{\text{Percentage change in price}} \times 100$$

Types of Elasticity of Supply

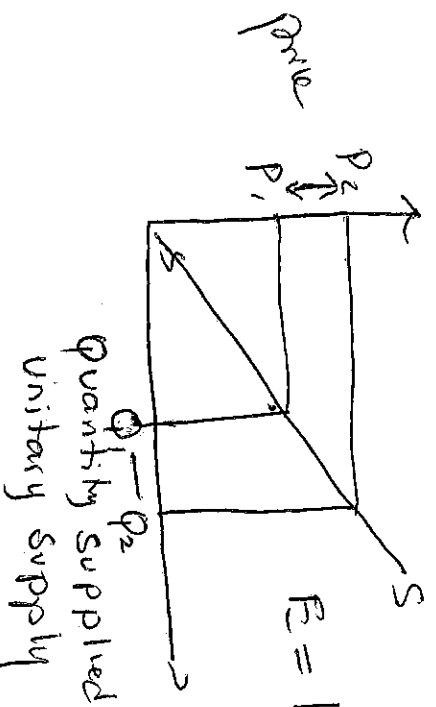
1. **ELASTIC Supply** - If a small change in price leads to a greater change in quantity supplied. It is greater than 1 or unitary. $E = \text{example}$



2. **INELASTIC Supply** - If a large change in price leads to a smaller or slight change in the quantity of goods supplied. ~~Formula~~ It is always less than one but greater than zero.



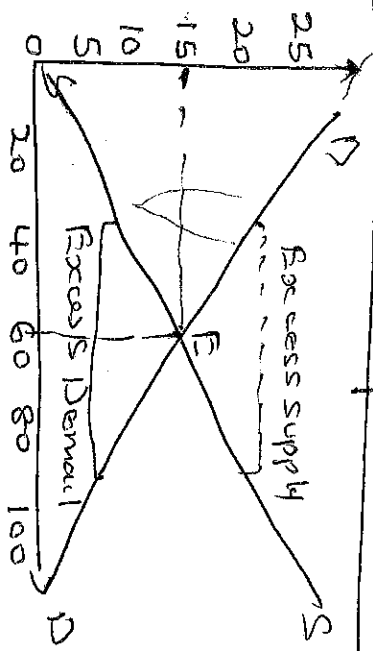
3. **UNITARY Supply** - when a change in price leads to an equal change in the quantity of goods supplied.



EQUILIBRIUM PRICE OF DEMAND AND SUPPLY (2)

Equilibrium price is that price at which the quantity demanded is equal to the quantity of commodities supplied. Example

Price	Quantity demanded	Quantity supplied
25	20	100
20	40	80
15	60	60
10	80	40
5	100	20



Quantity demanded and quantity supplied
Equilibrium price

From the above graph the equilibrium price is ₹15 and the quantity demanded is 60 as well as quantity supplied is 60. If the price increase to ₹25 producers will be willing to supply more which lead to excess supply. Similarly if the price is below to ₹5, consumers will be willing to demand more, which also lead to excess demand.

MATHEMATICAL APPROACH TO DEMAND AND SUPPLY FUNCTION

Example. Given that quantity demanded per period of time is a function of price expressed as $QD = 20 - 2P$ quantity supplied $QS = 6P - 12$. 1. Determine the equilibrium price and quantity bought and sold at that price. 2. If the price of the commodity is fixed at ₹6 what is the magnitude of the excess supply

Solution: $QD = 20 - 2P$
 $QS = 6P - 12$

① $QD = QS$
 $20 - 2P = 6P - 12$
 $-2P - 6P = -20 - 12$
 $-8P = 32$
 $8P = 32$
 $P = \frac{32}{8}$
 $P = 4.$

$QD = 20 - 2P$
 $= 20 - 2(4)$
 $= 12$

The magnitude of the excess supply will be $24 - 12 = 12$

② $QS = 6P - 12$
 $= 6(4) - 12$
 $= 24 - 12$
 $= 12.$

③ $QS = 6P - 12$
 $= 6(6) - 12$
 $= 36 - 12$
 $= 24$

Factors that Affect Elasticity of Supply ^X (5)

1. Availability of factors of production - supply will be elastic
2. The higher the production cost the more inelastic the supply
3. Durability of the product - The ~~durable~~ durable a product the elastic is the product.
4. Possibility of factor of substitution - If the input of production have more substitutes, the elastic will be the supply.
5. Production time - The shorter the period of production the more elastic the supply.

MEASUREMENT OF ELASTICITY OF SUPPLY

Formula: $ES = \frac{\text{Percentage Change in Supply}}{\text{Percentage Change in Price}} \times \frac{100}{1}$

Example: Ten (10) tubers of yam cost ₦10, 24 tubers were supplied and when the price increase ~~from~~ to ₦12, 30 tubers were supplied

1. Calculate the coefficient of price elasticity of supply
2. What type of elasticity.

Solution:

Price	Quantity supplied
10	24
12	30

1. Percentage change in Supply = $\frac{30 - 24}{24} \times \frac{100}{1} = \frac{6}{24} \times \frac{100}{1} = 25\%$

2. Percentage change in price = $\frac{12 - 10}{10} \times \frac{100}{1} = \frac{2}{10} \times \frac{100}{1} = 20\%$

3. Price elasticity of supply = $\frac{25\%}{20\%} = 1.25$

4. The coefficient of elasticity is elastic

Example 2. Price and quantity of a fish

Price ₦	Quantity supplied
10	20
14	24

1. Find the percentage change in quantity supplied.
2. Find the percentage change in price
3. Find the coefficient of elasticity
4. What type of elasticity

Solution

1. Percentage change in quantity supplied = $\frac{24 - 20}{20} \times \frac{100}{1} = \frac{4}{20} \times \frac{100}{1} = 20\%$

2. Percentage change in price = $\frac{14 - 10}{10} \times \frac{100}{1} = \frac{4}{10} \times \frac{100}{1} = 40\%$

3. Coefficient of elasticity $\hat{=}$ ES = $\frac{20\%}{40\%} = \frac{1}{2}$ or 0.5

4. Supply is inelastic.